

Chapter 6: Phase plane, $\dot{X} = F(X)$, $X \in \mathbb{R}^2$

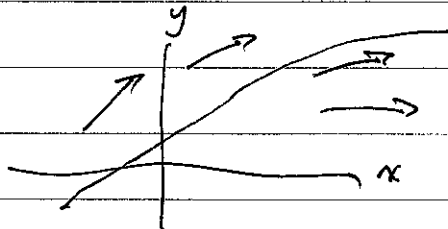
- Trajectories can't cross - limits how complicated dynamics can be
- "Nullclines" - organize the direction of vector fields
- Equilibria & their linear stability properties
- Hyperbolic equilibria & Hartman-Grobman Thm.
- Stable & unstable manifolds associated with hyperbolic equilibria

$$\left. \begin{aligned} \dot{x} &= F_1(x, y) \\ \dot{y} &= F_2(x, y) \end{aligned} \right\} \text{autonomous: } F_j \text{ don't depend on } t$$

$$X = \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$$

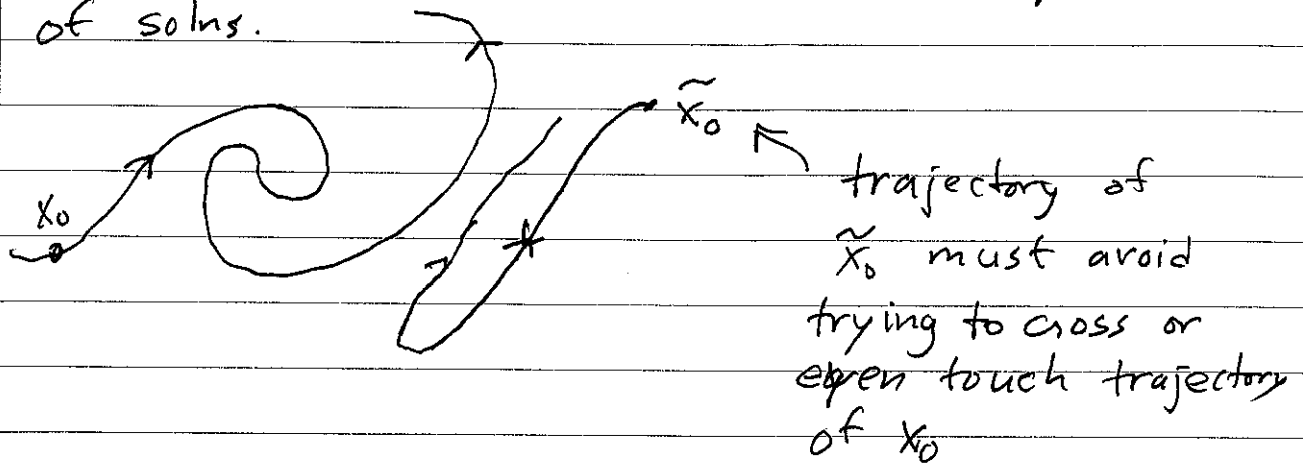
$$\dot{X} = F(X), \quad F = \begin{pmatrix} F_1 \\ F_2 \end{pmatrix}, \quad F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

assume F is at least C^1 : continuous with continuous partial derivatives $\Rightarrow$ existence uniqueness theorem holds.

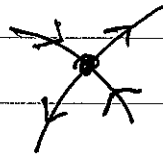


trajectories are everywhere tangent to the vector field.

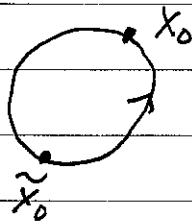
"Trajectories can't cross" due to uniqueness of solns.



don't confuse "crossing" with places where $F=0$



not a "crossing" an equilibrium



this can happen; the loop is a periodic soln. Trajectories associated with x_0 & $\tilde{x}_0$ coincide.

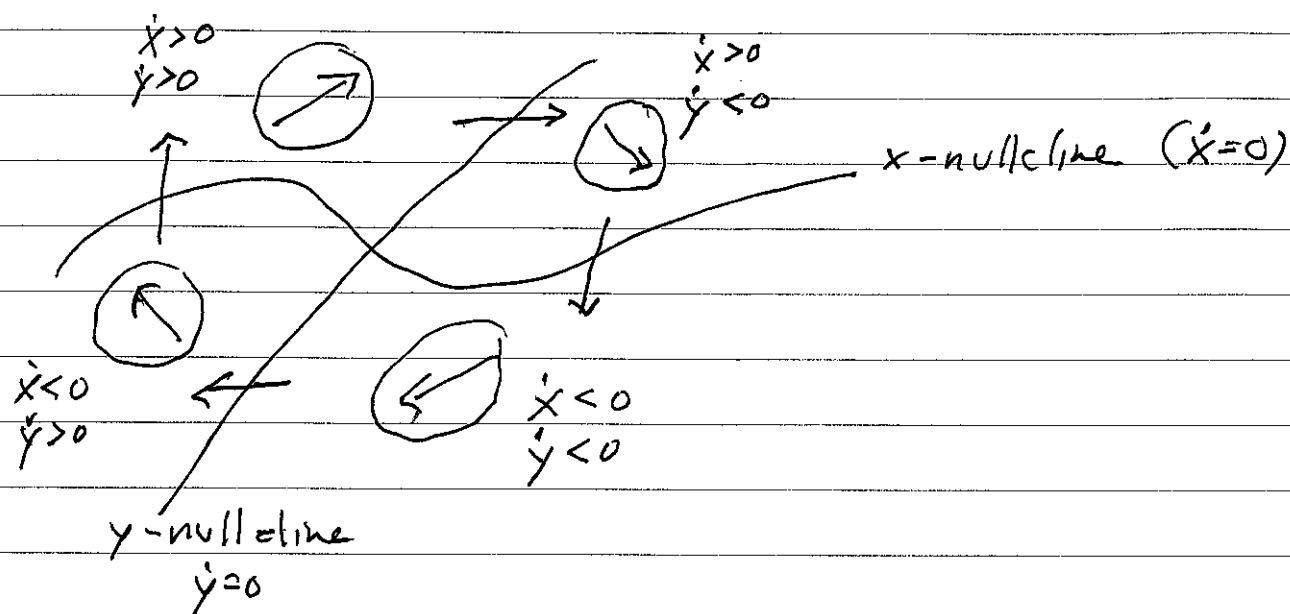
Simplest elements of phase portrait to determine

- Fixed pts / equilibria (x^*, y^*) where $F(x^*, y^*) = 0$
- Nullclines

$$\begin{aligned}\dot{x} &= F_1(x, y) \\ \dot{y} &= F_2(x, y)\end{aligned}$$

$$\begin{aligned}\text{x-nullcline} & F_1(x, y) = 0 \\ \text{y-nullcline} & F_2(x, y) = 0\end{aligned}$$

Intersections of nullclines - locations of fixed pts



Behavior near an equilibrium (x^*, y^*)

$$\left. \begin{aligned} \text{let } x &= x^* + u \\ y &= y^* + v \end{aligned} \right\} \quad \begin{aligned} |u|, |v| &\ll 1 \\ (\text{small perturbation}) \end{aligned}$$

$$F_1(x^*, y^*) = F_2(x^*, y^*) = 0$$

$$\begin{aligned} \dot{x} = \dot{u} &= F_1(x^* + u, y^* + v) = \frac{\partial F_1}{\partial x}(x^*, y^*) u + \frac{\partial F_1}{\partial y}(x^*, y^*) v + \dots \\ \dot{y} = \dot{v} &= F_2(x^* + u, y^* + v) = \frac{\partial F_2}{\partial x}(x^*, y^*) u + \frac{\partial F_2}{\partial y}(x^*, y^*) v + \dots \end{aligned}$$

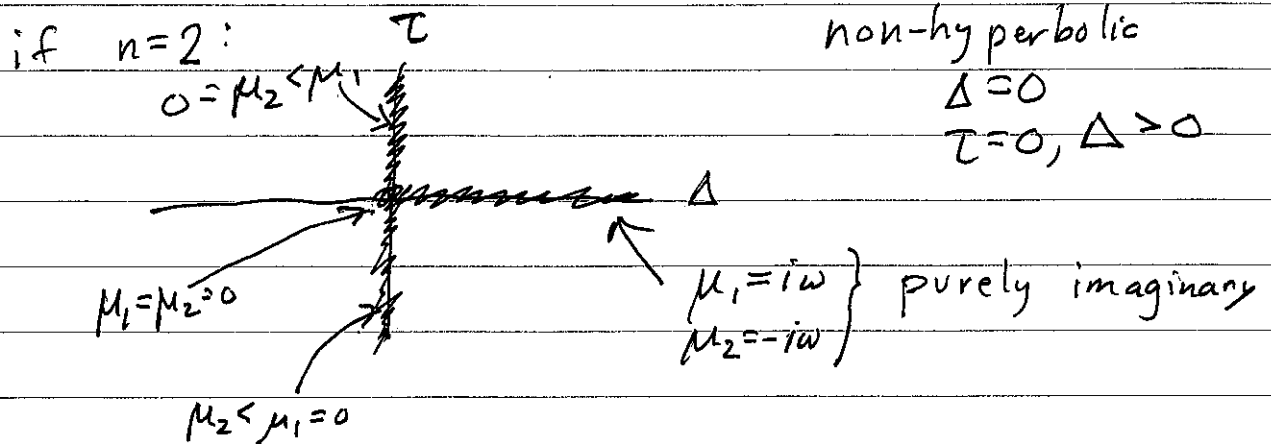
Linearized problem:

$$\begin{pmatrix} \dot{u} \\ \dot{v} \end{pmatrix} = \begin{pmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}$$

2×2 (constant)
 Jacobian
 matrix,
 denoted
 $DF(x^*, y^*)$

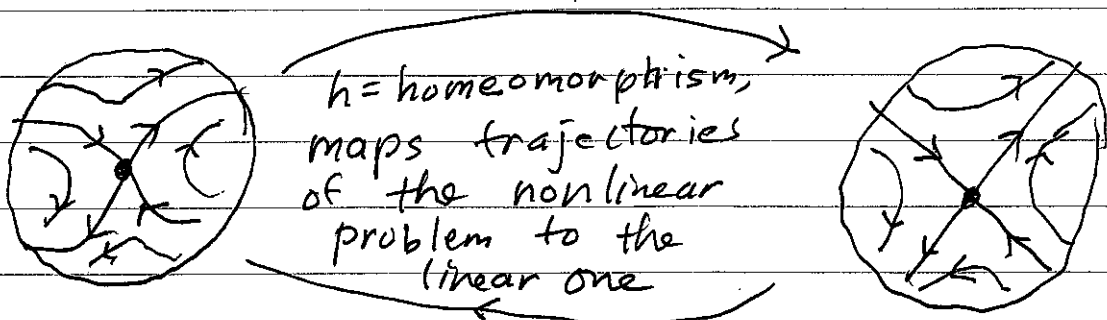
$(x, y) = (x^*, y^*)$

Def. Let x^* be an equilibrium soln. of $\dot{x} = F(x)$, $x \in \mathbb{R}^n$ (ie. $F(x^*) = 0$). We call x^* a hyperbolic fixed-pt if $\text{Re}(\mu) \neq 0$ for all eigenvalues μ of $DF(x^*)$.



Proved in more mathematical version of course:
 linearization determines stability properties of x^* if x^* is hyperbolic.

Hartman-Grobman Thm, is a picture, for hyperbolic fixed-pt.



$\dot{x} = F(x)$
 in neighborhood
 of x^*

$\dot{U} = DF(x^*) U$
 in neighborhood
 of $U = 0$

Implication of Hartman-Grobman

We can use linear stability calculations to determine stability of equilibria, provided the fixed-pt. is hyperbolic.

(If non-hyperbolic, linear stability can be inconclusive.)

linear asymptotic stability ($\operatorname{Re}(\mu) < 0, \forall \mu$)

$\Rightarrow$ asymptotic stability

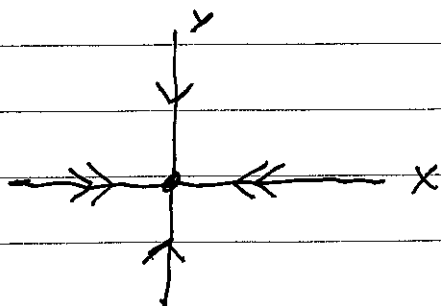
linear instability ($\exists \mu, \operatorname{Re}(\mu) > 0$)

$\Rightarrow$ instability

example:
$$\begin{cases} \dot{x} = -\lambda x + a x^2 + b x y \\ \dot{y} = -\mu y + c x^2 \end{cases} \quad \begin{matrix} \lambda > \mu > 0 \\ a, b, c \in \mathbb{R} \end{matrix}$$

$x=y=0$ is equilibrium for all parameters

linearized problem:
$$\begin{cases} \dot{x} = -\lambda x \\ \dot{y} = -\mu y \end{cases} \quad \begin{matrix} x = x_0 e^{-\lambda t} \quad \leftarrow \text{faster decay} \\ y = y_0 e^{-\mu t} \quad \leftarrow \text{slower decay} \end{matrix}$$



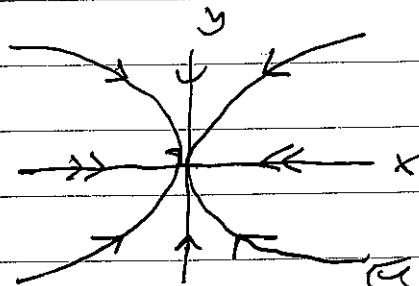
$(0,0)$ is hyperbolic
& asymptotically stable

Aside:
$$\begin{cases} \dot{x} = -\lambda x \\ \dot{y} = -\mu y \end{cases} \quad \frac{dx}{dy} = \frac{dx/dt}{dy/dt} = \frac{dx}{dt} \frac{dt}{dy} = \frac{\lambda}{\mu} \frac{x}{y}$$

$$\int_{x_0}^x \frac{1}{x} dx = \frac{\lambda}{\mu} \int_{y_0}^y \frac{1}{y} dy$$

$$\ln \left| \frac{x}{x_0} \right| = \frac{\lambda}{\mu} \ln \left| \frac{y}{y_0} \right| = \ln \left| \frac{y}{y_0} \right|^{\lambda/\mu}$$

$$\frac{x}{x_0} = \left(\frac{y}{y_0} \right)^{\lambda/\mu} \Rightarrow x = x_0 \left(\frac{y}{y_0} \right)^{\lambda/\mu} = C y^{(\lambda/\mu)}$$



$$\frac{\lambda}{\mu} > 1$$

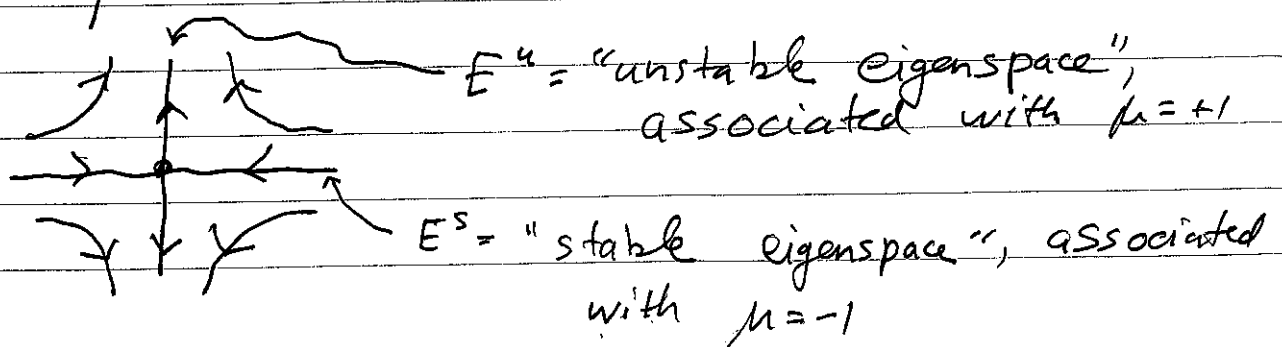
trajectories
come in
tangent to
slow direction.

Stable & unstable manifolds of hyperbolic fixed pts.

Warm-up:

Linear problem in $\mathbb{R}^2$ with a saddle pts

$$\begin{aligned} \dot{x} &= -x \\ \dot{y} &= y \end{aligned} \quad \left(\text{vs.} \quad \begin{aligned} \dot{x} &= -x + 4x^2 - 3xy \\ \dot{y} &= y + 7x^2 \end{aligned} \right)$$



If $X_0 \in E^s$, then $X(t) \in E^s$, $\lim_{t \rightarrow \infty} X(t) = 0$

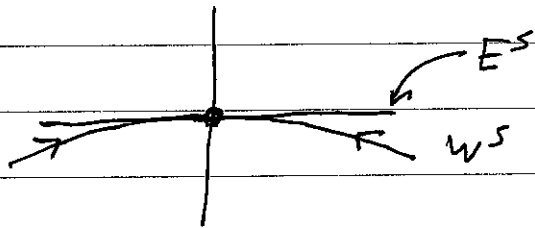
If $X_0 \in E^u$, then $X(t) \in E^u$, $\lim_{t \rightarrow -\infty} X(t) = 0$

Nonlinear problem: replace stable/unstable eigenspaces of origin by stable/unstable manifolds of fixed-pt X^*

If $X_0 \in W^s$, then $X(t) \in W^s$, $\lim_{t \rightarrow \infty} X(t) = X^*$

If $X_0 \in W^u$, then $X(t) \in W^u$, $\lim_{t \rightarrow -\infty} X(t) = X^*$

Invariant stable (or unstable) manifolds



- invariant : if $X_0 \in W^s \Rightarrow X(t) \in W^s$
- It's tangent to the stable eigenspace at X^*
- near $X = X^*$ it can be represented as a graph over the stable eigenspace.